ASSIGNMENT-1

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course Name : Fundamentals of Machine learning

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1) Task-1

Identify the type of Machine learning system for each application.

(a) Loan default prediction

Supervised learning – classification. It predicts whether a customer will default or not using labelled data.

(b) Customer segmentation

unsupervised learning – clustering. It groups customers based on similar characteristics

(c) Credit spend forecasting

Supervised learning – Regression.
It learns the best refill decisions using rewards and historical data

(d) ATM cash Refill policy

Reinforcement learning. It learns the best refill the decisions using rewards and penalties

(e) Fraud detection using with 2% labelled data

Semi Supervised learning. It uses a small the amount of labelled data along with unlabelled data

for fraud detection has 2% labelled data in the supervised learning

Task 2:

(2) calculate the probability of disease using Baye's theorem

Given,

Disease prevalence = 2%.

$$= \frac{2}{100} \times 100$$

$$= 0.02$$

Sensitivity = 95% = 0.95%.

Specificity = 90% = 0.90

False positive rate = 1 − 0.90 = 0.10

using Baye's theorem:

$$P(D|+) = \frac{P(+|D)\,P(D)}{P(+|D)\,P(D) + P(+|\bar{D})}$$

Substituting,

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.117} \times 1$$

$$= 0.1624$$

$$\boxed{P(D|+) = 16.24\%}$$

Therefore, the probability that a person actually has the disease after reciving a +ve test result 16.24%

Task-3: JOINT pro

X

0

1

2

a) M

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Task-3:

JOINT probability distribution

Given the Joint probability table:

X	Y=0	Y=1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

a) Marginal probabilities

for X: $P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

For Y: $P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$

$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$

(b) Expected values:

$E(X) = 0(0.30) + 1(0.40) + 2(0.30)$

$\boxed{E(X) = 1}$

$E(Y) = 0(0.45) + 1(0.55)$

$\boxed{E(Y) = 0.55}$

(c) variance of X

$E(X^2) = 0^2(0.30) + 1^2(0.40) + 2^2(0.30)$

$= 1.60$

$var(X) = E(X^2) - [E(X)]^2$

$= 1.60 - 1^2$

$= 0.60$

$\boxed{var(X) = 0.60}$

(d) conditional probability

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

$$= \frac{0.10}{0.55}$$

$$\boxed{P(X=2|Y=1) = 0.1818}$$

(e) covariance

$$\text{cov}(X,Y) = 0.45 - (1 \times 0.55)$$

$$= -0.10$$

(f) correlation coefficient

$$\text{var}(Y) = 0.55(0.45) = 0.2475$$

$$\rho = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\,\text{var}(Y)}}$$

$$= \sqrt{0.60 \times 0.2475}$$

$$\boxed{\rho = -0.2593}$$

(e) Independence

For independent variables

$$P(X=0, Y=0) = P(X=0)\,P(Y=0)$$

But,

$$0.10 \neq 0.30 \times 0.45$$

$$= 0.135$$

Therefore, X and Y are not Independent

$$\boxed{\text{Independence} = 0.135}$$

Task 4:

Normal Distr

Given,

X ~ N

Mean =

Standa

a) Prob

Task 4:

Normal Distribution and Maximum Likelihood Estimation

Given,

$X \sim N(50, 16)$

Mean = 50

Standard deviation = 4

a) probability that X is greater than 58

$$Z = \frac{58-50}{4}$$

$$\boxed{Z = 2}$$

$$P(X > 58) = P(Z < 2)$$

$$\boxed{P(X > 58) = 0.0228}$$

b) probability that X lies between 46 and 54

$$Z_1 = \frac{46-50}{4}$$

$$\boxed{Z_1 = -1}$$

$$Z_2 = \frac{54-50}{4}$$

$$\boxed{Z_2 = 1}$$

$$P(46 < X < 54) = P(-1 < Z < 1)$$

$$\boxed{P = 0.6826}$$

(c) sensor readings

Given readings: 12, 15, 11, 14, 13

$$\text{Mean } \bar{x} = \frac{12+15+11+14+13}{5}$$

$$\boxed{\bar{x} = 13}$$

Squared deviations:

$(12-13)^2 + (15-13)^2 + (11-13)^2 + (14-13)^2 + (13-13)^2$

$= 10$

MLE variance:

$\sigma^2_{MLE} = 10/5$

$\boxed{\sigma^2_{MLE} = 2}$

unbiased simple variance:

$s^2 = \frac{10}{5-1}$

$\boxed{s^2 = 2.5}$

Tasks:

5) Entropy and coding

Given probabilities

$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$

(a) Entropy

$H(X) = -\sum P(x) \log_2 P(x)$

$= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$

$= 0.5 + 0.5 + 0.375 + 0.375$

$= 1.75$ bits

$\boxed{H(X) = 1.75 \text{ bits}}$

(b) Uniform distribution energy

$H(X) = \log_2 n$

$= 109.4$

$\boxed{H(X) = 2 \text{ bits}}$

c) calculate redundancy

$R = H_{max} - H(X)$

$R = 2 - 1.75$

$\boxed{R = 0.25 \text{ bits}}$

Percentage:

$= \frac{0.25}{2} \times 100$

$= 12.5\%$

d) calculate average code length

Given codes: 0, 10, 110, 111

code lengths: 1, 2, 3, 3

$L = \Sigma P(x) l(x)$

$L = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$

$L = 0.5 + 0.5 + 0.375 + 0.375$

$\boxed{L = 1.75 \text{ bits}}$

e) Check optimality

$L = H(X)$

$\boxed{L = 1.75}$

Therefore, the code is optimal.

Task 6:

6) Kullback- Leibler Divergence

Given,

$P = (0.5, 0.25, 0.25)$

$Q = (1/3, 1/3, 1/3)$

a) calculate $D_{KL}(P||Q)$

$$D_{KL}P(||Q) = \Sigma P(x) \log_2 \frac{P(x)}{Q(x)}$$

$$= 0.5 \log_2 \frac{0.5}{1/3} + 0.25 \log_2 \frac{0.25}{1/3} + 0.25 \log_2 \frac{0.25}{1/3}$$

$$= 0.5 \log_2 (1.5) + 0.5 \log_2 (0.75)$$

$$= 0.29248 - 0.20752$$

3 $\boxed{D_{KL}(P||Q) = 0.08496 \text{ bits}}$

b) calculate $D_{KL}(Q||P)$

$$D_{KL}(Q||P) = \Sigma Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$= \frac{1}{3} \log_2 \frac{1/3}{0.5} + \frac{1}{3} \log_2 \frac{1/3}{0.25} + \frac{1}{3} \log_2 \frac{1/3}{0.25}$$

$$= \frac{1}{3} \log_2 (0.667) + \frac{2}{3} \log_2 (1.333)$$

$\boxed{D_{KL}(Q||P) = 0.08170 \text{ bits}}$

(c) Assymetry

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

Hence, KL divergence is asymmetric

Task 7:

7) Entropy and Mutual information

Given that,

A	B	probability
0	0	0.4
0	1	0.1
1	0	0.1
1	1	0.4

a) Entropy of A

$P(A=0) = 0.5, \quad P(A=1) = 0.5$

$H(A) = -[0.5 \log_2 0.5 + 0.5 \log_2 .05]$

$\boxed{H(A) = 1 \text{ bit}}$

b) Entropy of B

$P(B=0) = 0.5,$

$P(B=1) = 0.5$

$\boxed{H(B) = 1 \text{ bit}}$

c) Joint entropy

$H(A,B) = -\sum P(A,B) \log_2 P(A,B)$

$= -[0.4 \log_2 0.4 + 0.1 \log_2 0.1 + 0.1 \log_2 0.1 + 0.4 \log_2 0.4]$

$\boxed{H(A,B) = 1.7219 \text{ bits}}$

d) conditional entropy

$H(A|B) = H(A,B) - H(B)$

$H(A|B) = 1.7219 - 1$

$\boxed{H(A|B) = 0.7219 \text{ bits}}$

e) Mutual information

$I(A;B) = H(A) + H(B) - H(A,B)$

$I(A;B) = 1+1-1.7219$

$\boxed{I(A;B) = 0.2781 \text{ bits}}$

f) feauture interpretation

Mutual information measures the information shared between two variables.

8) **Task 8:** Bayesian classification using Gaussian distribu

Given,

$\mu_1 = 2, \mu_2 = 6$

$\sigma^2 = 4$

$P(\omega_1) = 0.6, P(\omega_2) = 0.4$

a) Find decision threshold

$$x^* = \frac{\mu_1+\mu_2}{2} + \frac{\sigma^2}{\mu_2-\mu_1} \ln \frac{P(\omega_1)}{P(\omega_2)}$$

$$= \frac{2+6}{2} + \frac{4}{4} \ln \frac{0.6}{0.4}$$

$$= 4 + \ln(1.5)$$

$$= 4 + 0.4055$$

$\boxed{x^* = 4.4055}$

b) Classify $x = 4.5$

$4.5 > 4.4055$

$\boxed{\text{predicted class} = \omega_2}$

c) calculate classification error

$P(\text{error}) = 0.6P(z > 1.2027) + 0.4P(z < -0.7973)$

from the z-table

$= 0.6(0.$

$= 0.066$

P(error)

d) effe

If P

a)

From the z-table:

$= 0.6(0.1146) + 0.4(0.2126)$

$= 0.066876 + 0.08504$

$P(\text{error}) = 0.1538 = 15.38\%$

d) effect of increasing prior probability

If $P(w_1)$ increases, the threshold moves towards the right

Task 9:

9) Minimum Risk classification

Given,

$P(D|x) = 0.3$, $P(H|x) = 0.7$

Decision	Disease	Healthy
predict disease	0	8
predict healthy	50	0

a) Risk of predicting disease

$R(\alpha_1) = 0(0.3) + 8(0.7)$

$= 0 + 5.6$

$\boxed{R(\alpha_1) = 5.6}$

b) Risk of predicting healthy

$R(\alpha_2) = 50(0.3) + 0(0.7)$

$= 15 + 0$

$\boxed{R(\alpha_2) = 15}$

c) Minimum risk detection

$5.6 < 15$

$\boxed{\text{Predict disease}}$

d) compare with MAP

$P(D|x) = 0.3$, $P(H|x) = 0.7$

MAP selects the highest probability.

MAP decision = Healthy

e) calculate posterior threshold

Let $p = P(D|x)$

$8(1-p) < 5Op$

$8 - 8p < 5Op$

$8 < 58p$

$p > 8/58$

$P > 0.1379 = 13.79\%$

Task 10:

10) Bias- variance Trade-off and model selection

Given,

Degree	Training MSE	validation MSE
1	9.0	9.5
2	4.1	4.6
3	2.2	2.9
5	0.9	3.6
9	0.1	11.8

a) Calculate generalization gap

Gap = validation MSE - Training MSE

Degree1: $9.5 - 9.0 = 0.5$

Degree2: $4.6 - 4.1 = 0.5$

Degree 3 : 2.9 - 2.2 = 0.7
Degree 5 : 3.6 - 0.9 = 2.7
Degree 9 : 11.8 - 0.1 = 11.7

Final table :

Degree	Generalization gap
1	0.5
2	0.5
3	0.7
5	2.7
9	11.7

b) Bias - Variance trade - off

Low degree models may have high bias
High degree models may have high variance

c) Select the best model

Minimum validation MSE
= 2.9

Best degree = 3

d) Why does degree 9 overfit ?

Training MSE = 0.1
Validation MSE = 11.8

The large difference indicates overfitting

e) Two improvements

1) Apply Ridge or Lasso regularization
2) Reduce polynomial degree or select improtant features.

Task II:

ii) Bayesian Estimation of coin Bias given

Given,

prior = Beta(2,2)

a) For 10 tosses with 7 heads

Number of tails:

$10 - 7 = 3$

MLE:

$\hat{p} = 7/10$

MLE = 0.7

posterior distribution:

Beta(2+7, 2+3) = Beta(9,5)

MAP: $\hat{p}_{MAP} = \frac{\alpha - 1}{\alpha + \beta - 2}$

$= \frac{9-1}{9+5-2}$

$= 8/12$

MAP = 0.6667

posterior mean:

$E(P) = \frac{\alpha}{\alpha + \beta}$

$= \frac{9}{9+5}$

posterior mean = 0.6429

b) For 100 tosses with 70 heads

Number of tails: $100 - 70 = 30$

as given

MLE : 70/100

MLE = 0.7

Posterior distribution:

Beta (2+70, 2+30)

= Beta (72, 32)

MAP :

$$\frac{72-1}{72+32-2}$$

= 71/102

MAP = 0.6961

Posterior mean: $= \frac{72}{72+32}$

= 72/104

posterior mean = 0.6923

Task 12 :

12) sepsis Early warning system

Given,

Total = 1000

prevalence = 5%.

sensitivity = 90%.

specificity = 80%.

a) calculate confusion matrix

Now,

Actual sepsis patients,

$$1000 \times \frac{5}{100}$$

$$= 50$$

Healthy patients,

$$1000 - 50$$

$$= 950$$

True positive;

$$TP = 50 \times 0.90$$

$$\boxed{TP = 45}$$

False negative;

$$FN = 50 - 45$$

$$\boxed{FN = 5}$$

True negative:

$$TN = 950 \times 0.80$$

$$\boxed{TN = 760}$$

False positive;

$$FP = 950 - 760$$

$$\boxed{FP = 190}$$

S.Name	predicted sepsis	predicted Healthy
Actual sepsis	45	5
Actual Healthy	190	760

b) calculate PPV

PPV =

b) calculate PPV

$$PPV = \frac{TP}{TP+FP}$$

$$= \frac{45}{45+190}$$

$$= \frac{45}{235}$$

PPV = 19.15%

c) Calculate NPV

$$NPV = \frac{TN}{TN+FN}$$

$$= \frac{760}{760+5}$$

$$= \frac{760}{765}$$

NPV = 93.35%

d) Calculate cost comparison

Given,

False Negative cost = ₹100

False positive cost = ₹5

Model based system

$$cost = (FN \times 100) + (FP \times 5)$$

$$= (5 \times 100) + (190 \times 5)$$

$$= 500 + 950$$

Cost = ₹1450

Alert all patients:

cost = 950 × 5

= ₹4750

Never alert:

cost = 50 × 100

= ₹5000

e) calculate Bayes decision threshold

let $p = P(\text{Disease}/x)$

$R(\text{Alert}) = 5(1-p)$

$R(\text{No Alert}) = 100p$

For alerting,

$5(1-p) < 100p$

$5 - 5p < 100p$

$5 < 105p$

$p > 5/105$

$p > 0.0476$

$p > 4.76\%$

$P > 0.0476 = 4.76\%$

∴ The model based system has the lowest cost of ₹1450 //